

Raging Bits ESP32 WS2812 8 CHAR DISPLAY v1.1

Top level specs

ESP32-WROOM-16Mbit central Module

8 char display with full RGB 5x7 matrix digit (ws2812 leds)

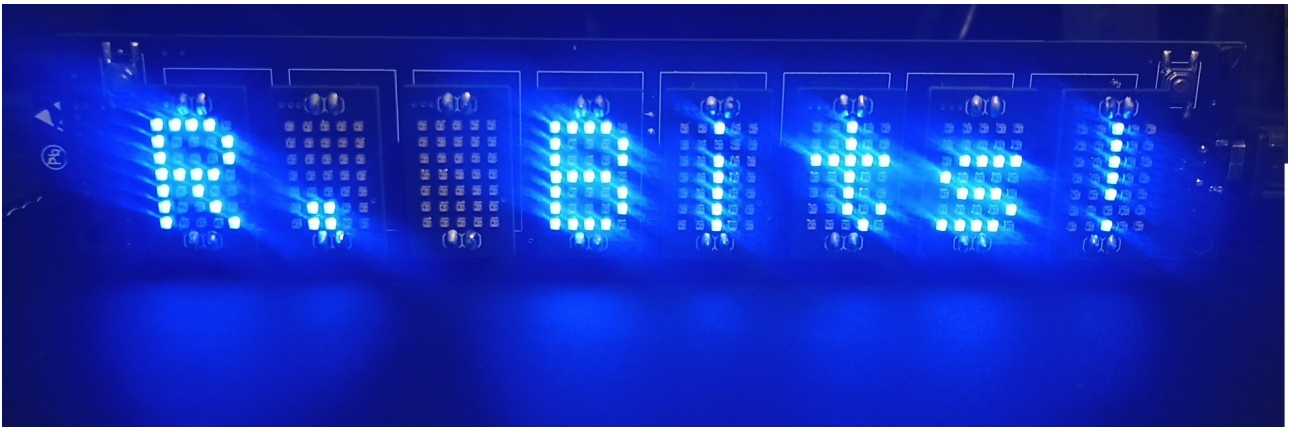
Battery backed up RTC. (CR1220 coin cell)

1 configurable Led.

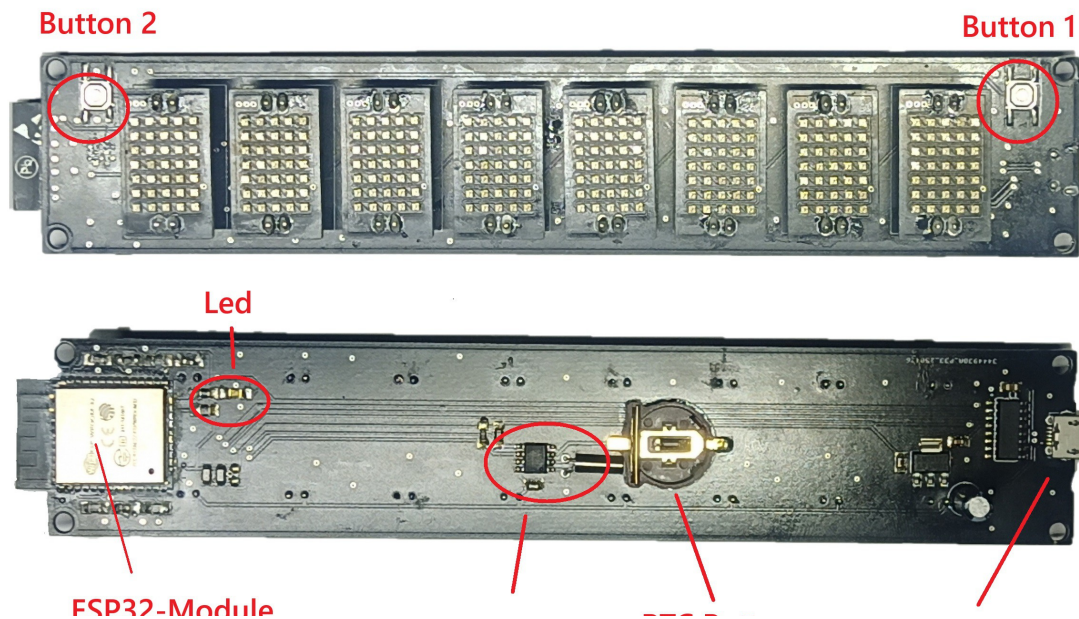
1 Serial Port (fixed 115200bps)

2 buttons for clock control interface

Standard micro USB 5V power serial port CH430.



Device interface



Compatible with WLED and many other softwares.

Hardware

Power

Power consumption at 5V is kept within USB standard, by limiting the brightness.

ESP32 Pinout

WS2812 control:	Pin 27 - GPIO16
Button2:	Pin 37 - GPIO23
Button1:	Pin 10 - GPIO25
Info LED:	Pin 24 - GPIO2
I2C SDA:	Pin 33 - GPIO21
I2C SCL:	Pin 36 - GPIO22

Programmer

The ESP is programmed directly from serial port.
The Port is fully wired oriented towards ESP tools programmers.

ESP32 programmer serial port

TX: Pin 35 - GPIO1

RX: Pin 34 - GPIO3
RTS: Pin 3 – ENABLE (Reset)
DTR: Pin 25 – GPIO0 (Boot)

Uart

For the Default application, the UART is fixed at 115200bps, 1 start bit, 1 stop bit, no flow control and no parity.

INFO LED

A simple LED can be set or reset by serial commands.

Buttons

The buttons are used to directly interface and make use of the display as a clock/calendar.

Button 1 click

A button 1 click will simply move the text to another colour.

There are 11 preset colours to choose from:

- Rainbow - keeps the text moving through all the colours.

- Red - solid

- Orange - solid

- Yellow - solid

- Green - solid

- Light Cyan - solid

- Cyan - solid

- Blue - solid

- Purple - solid

- Pink - solid

- White - solid

Button 2 click

A button 2 click will simply move through the clock date display options.

There are 4 different options available:

- Only time – No rotation.

- Only date – Rotating.

- Time and date – Rotating.

- Alternate Time and date – No rotation for time for 45s. Rotating date for 15s.

Button 1 hold+ Button 2 hold

When holding both buttons for over a second, the setup menu is invoked.

Once active, if no button presses are detected within 30s, the setup exits back to normal clock.

Setup Button 2

While in setup, button 2 controls values by incrementing by one unit with a click or continuously by holding.

Setup Button 1

While in setup, button 1 controls the settings menu items by incrementing by one unit with a click or continuously by holding.

Setup menu

The setup menu has 7 items and always goes top to bottom once before exiting:

Set year – (2023 to 2099).

Set month – (1 to 12).

Set day – (1 to 31).

Set hour – (0 to 23).

Set minute – (0 to 59).

Set brightness – (10 levels).

Save – (yes or no).

Software

Serial port Commands

As mentioned before, the Default software allows for formatted commands at 115200bps. The system is designed to have the commands synchronized using a simple framing system. The frame starts with a Line Feed character, followed by the command identification, the command data if any, and ends with a Carriage Return. This means all command data will be in ASCII not binary.

[`\n`] [Command ID] [Data (if any)] [`\r`]

Show Text – '01' (0x00 0x31)

Every new text entry resets the display rotation point. Due to the nature of the HPDL1414, only a limited set of characters are supported, Numbers, only Upper Case letters and some signs. (There is a table with the full supported contents in the Notes.)

[`\n`] [01] [Text to show] [`\r`]

```
Serial > n01HELLO WORLD    \r
Serial < \nOK\r
```

Display now rotates the message "HELLO WORLD ".

Show Time – '02' (0x00 0x32)

[`\n`] [02] [`\r`]

```
Serial > n02\r
Serial < \nOK\r
```

Display will now show the time.

Show Time Date – '03' (0x00 0x33)

[`\n`] [03] [`\r`]

```
Serial > n03\r
Serial < \nOK\r
```

Display will now show the time and date rotating.

Get Time-Date – '04' (0x00 0x34)

[`\n`] [04] [`\r`]

```
Serial > n04\r
```

Serial < \n00:10:13 01 FEB 2023\r

Display will return the time and date.

Set Time-Date – '05' (0x00 0x35)

The time and date format MUST BE EXACTLY as follow.

hh:mm:ss dd/MM/yyyy

where:

hh – 2 digit hours
mm – 2 digit minutes
ss – 2 digit seconds
dd – 2 digit day
MM – 2 digit month
yyy – 4 digit year

[\n] [05] [hh:mm:ss dd/MM/yyyy] [\r]

Serial > \n0500:10:13 01/02/23\r

Serial < \nOK\r

Display will return the time and date.

Set Work LED – '06' (0x00 0x36)

[\n] [06] [Status] [\r]

Serial > \n060\r

Serial < \nOK\r

Led is now OFF.

Serial > \n81\r

Serial < \nOK\r

Led is now ON.

Set Display text colour – '07' (0x00 0x37)

Sets the text colour with either a solid colour or a progressive rainbow.

The rainbow effect is chosen by having the effect data as 0.

The solid colour is chose nby having the effect data as 1 and will admit a colour setup of a combination of 10 gradients per component of the RGB.

[\n] [07] [(0/1)Effect] [R(0-9)] [G(0-9)] [B(0-9)] [\r]

Serial > \n070000\r

Serial < \nOK\r

Sets Rainbow colours.

Serial > \n071999\r
Serial < \nOK\r

Serial > \n071555\r
Serial < \nOK\r

Led Solid white

Serial > \n071900\r
Serial < \nOK\r

Led Solid red

Set Display text brightness – '08' (0x00 0x38)

The text brightness command will set the intensity of the display brightness. A simple factor over the colour. The brightness can be set between 0 and 100.

[\\n] [08] [(0-1)] [(0-9)] [(0-9)] [\\r]

Serial > \n08000\r
Serial < \nOK\r

Brightness set to zero

Serial > \n08058\r
Serial < \nOK\r

Brightness set to 58

Set Wifi – '09' (0x00 0x39)

The text wifi command will set the wifi SSID and Password to the clock. This way the clock will be able to connect to it and sync time automatically .

[\\n] [09] [SSID] [(space)] [PASSWORD] [\\r]

Serial > \n09MyWifiName MyWifiPassword\r
Serial < \nOK\r

Set NTP server – '10' (0x31 0x30)

The text set NTP server command will set the NTP IP or URL to connect to for the automatic time sync.

```
[\\n] [10] [ntp server] [\\r]
```

```
Serial > \\n10192.168.1.25\\r  
Serial < \\nOK\\r
```

```
Serial > \\n10pool.ntp.org\\r  
Serial < \\nOK\\r
```

Set time zone – '11' (0x31 0x31)

The text set timezone command will set the respective timezone needed for the correct time to be set over the automatic time sync. The command is index based for pre recorded timezones off the table, but can also use custom timezones. The custom timezones use **POSIX format string**.

The current available table timezones are:

WESTERN HEMISPHERE (UTC-12 to UTC-1)	
case 0: return "AoE12";	// Baker Island (UTC-12)
case 1: return "NUT11";	// American Samoa / Niue (UTC-11)
case 2: return "HST10";	// Hawaii / Honolulu (UTC-10, No DST)
case 3: return "HDT9:30";	// Marquesas Islands (UTC-9:30, No DST)
case 4: return "AKST9AKDT,M3.2.0,M11.1.0";	// Alaska / Anchorage
case 5: return "PST8PDT,M3.2.0,M11.1.0";	// Pacific Time (Los Angeles / Vancouver)
case 6: return "MST7";	// US Arizona (Mountain Time, No DST)
case 7: return "MST7MDT,M3.2.0,M11.1.0";	// Mountain Time (Denver / Edmonton)
case 8: return "CST6CDT,M3.2.0,M11.1.0";	// Central Time (Chicago / Winnipeg)
case 9: return "CST6CDT,M4.1.0,M10.5.0";	// Mexico (Central / Mexico City)
case 10: return "EST5EDT,M3.2.0,M11.1.0";	// Eastern Time (New York / Toronto)
case 11: return "AST4";	// Caribbean / Puerto Rico (Atlantic No DST)
case 12: return "AST4ADT,M3.2.0,M11.1.0";	// Atlantic Time (Halifax / Canada)
case 13: return "NST3:30NDT,M3.2.0,M11.1.0";	// Newfoundland (UTC-3:30 + DST)
case 14: return "<-03>3";	// Argentina (Buenos Aires)
case 15: return "<-03>3<-02>,M10.2.0/0,M3.2.0/0";	// Chile (Santiago - Southern Hemisphere DST)
case 16: return "WGST3WGT,M3.2.0,M11.1.0";	// Western Greenland
case 17: return "<-01>1";	// Cape Verde (UTC-1)

BRAZIL SUB-DATABASE (Strictly mapped by territory)

case 18: return "<-02>2"; // Brazil (Fernando de Noronha, UTC-2)
case 19: return "<-03>3"; // Brazil (Official Brasília / São Paulo / Rio, UTC-3)
case 20: return "<-04>4"; // Brazil (Manaus / Amazon, UTC-4)
case 21: return "<-05>5"; // Brazil (Rio Branco / Acre, UTC-5)

EUROPE, UK & AFRICA (UTC 0 to UTC+3)

case 22: return "GMT0"; // Iceland / Reykjavik (UTC 0, No DST)
case 23: return "GMT0BST,M3.5.0/1,M10.5.0"; // UK / London / Dublin / Lisbon (GMT/BST)
case 24: return "WAT-1"; // West Africa (Lagos / Kinshasa, UTC+1)
case 25: return "CET-1CEST,M3.5.0,M10.5.0/3"; // Central Europe (Paris / Berlin / Rome / Madrid)
case 26: return "CAT-2"; // Central Africa (Johannesburg, UTC+2)
case 27: return "EET-2EEST,M3.5.0,M10.5.0/3"; // Eastern Europe (Athens / Helsinki / Kyiv)
case 28: return "EET-2EEST,M4.lastFriday,M10.lastThursday/23"; // Egypt Standard Time (Dynamic Africa DST)
case 29: return "IST-2IDT,M3.4.4/26,M10.5.0"; // Israel Standard Time
case 30: return "AST-3"; // East Africa / Saudi Arabia / Nairobi (UTC+3)
case 31: return "MSK-3"; // Moscow / St. Petersburg (UTC+3 Fixed)

MIDDLE EAST & ASIA (UTC+3:30 to UTC+9)

case 32: return "IRST-3:30IRDT,M3.3.0/0,M9.3.0/0"; // Iran / Tehran (Fractional + Custom Rules)
case 33: return "GST-4"; // Gulf / Dubai / Baku (UTC+4)
case 34: return "AFT-4:30"; // Afghanistan / Kabul (UTC+4:30)
case 35: return "UZT-5"; // Pakistan / Karachi / Uzbekistan (UTC+5)
case 36: return "IST-5:30"; // India / New Delhi / Sri Lanka (UTC+5:30)
case 37: return "NPT-5:45"; // Nepal / Kathmandu (UTC+5:45)
case 38: return "BST-6"; // Bangladesh / Dhaka / Omsk (UTC+6)
case 39: return "MMT-6:30"; // Myanmar / Yangon (UTC+6:30)
case 40: return "ICT-7"; // Indochina / Bangkok / Jakarta / Vietnam (UTC+7)
case 41: return "CST-8"; // China / Beijing / Singapore / Taipei / Perth (UTC+8)
case 42: return "ACWST-8:45"; // Eucla, Western Australia (UTC+8:45)
case 43: return "JST-9"; // Japan / Tokyo / Seoul (UTC+9)

OCEANIA & AUSTRALIA (UTC+9:30 to UTC+14)

case 44: return "ACST-9:30ACDT,M10.1.0,M4.1.0/3"; // Central Australia / Adelaide (Flipped DST)
case 45: return "AEST-10"; // Queensland / Brisbane (UTC+10, No DST)

case 46: return "AEST-10AEDT,M10.1.0,M4.1.0/3"; // Eastern Australia / Sydney /
 Melbourne (Flipped DST)
 case 47: return "LHST-9:30LHDT,M3.2.0/2,M11.1.0/2"; // Lord Howe Island (30-min DST
 shift variation)
 case 48: return "SBT-11"; // Solomon Islands / Vladivostok (UTC+11)
 case 49: return "NZST-12NZDT,M9.5.0,M4.1.0/3"; // New Zealand / Auckland (Flipped
 DST)
 case 50: return "CHAST-11:45CHADT,M9.5.0,M4.1.0/3"; // Chatham Islands, NZ
 (UTC+12:45 Fractional + DST)
 case 51: return "TOT-13"; // Tonga / Apia / Samoa (UTC+13)
 case 52: return "LINT-14"; // Line Islands / Kiritimati (UTC+14)

ADVANCED COMPREHENSIVENESS EXTENSION LAYER

case 53: return "EST5"; // Cancun, Mexico (Eastern Standard, No DST)
 case 54: return "AST4ADT,M3.2.0/0,M11.1.0/0"; // Bermuda (Atlantic Standard + DST
 Variant)
 case 55: return "<-03>3<-02>,M10.1.0/0,M3.2.0/0"; // Paraguay / Asunción (Flipped
 Southern DST)
 case 56: return "<-04>4<-03>,M10.2.0/0,M5.2.0/0"; // Chile - Easter Island Variant
 case 57: return "AZOT1AZOST,M3.5.0/0,M10.5.0/1"; // Azores, Portugal (UTC-1 +
 DST)
 case 58: return "EGT1EGST,M3.5.0/0,M10.5.0/1"; // Eastern Greenland /
 Ittoqqortoormiit
 case 59: return "WET0WEST,M3.5.0/1,M10.5.0/2"; // Western European Time (Canary
 Islands, Madeira)
 case 60: return "EET-2EEST,M3.5.0/3,M10.5.0/4"; // Moldova / Chisinau (Eastern
 Europe Variant)
 case 61: return "EET-2EEST,M3.5.0/1,M10.5.0/2"; // Lebanon / Beirut (Eastern Europe
 Variant)
 case 62: return "AST-3ADT,M3.5.0/0,M10.5.0/0"; // Jordan & Syria (Fixed Dynamic
 Offset Rule)
 case 63: return "MUT-4"; // Mauritius / Seychelles (UTC+4, No DST)
 case 64: return "ORAT-5"; // Oral / Western Kazakhstan (UTC+5)
 case 65: return "ALMT-5"; // Almaty / Eastern Kazakhstan / Kyrgyzstan
 (UTC+5)
 case 66: return "TJT-5"; // Tajikistan / Turkmenistan (UTC+5)
 case 67: return "YEKST-5"; // Yekaterinburg, Russia (UTC+5 Fixed)
 case 68: return "OMST-6"; // Novosibirsk / Krasnoyarsk / Central Asia
 (UTC+6)
 case 69: return "HOVT-7"; // Western Mongolia / Hovd (UTC+7)
 case 70: return "ULAT-8"; // Central Mongolia / Ulaanbaatar (UTC+8)
 case 71: return "TLT-9"; // East Timor / Dili (UTC+9)
 case 72: return "ACST-9:30"; // Northern Territory, Australia / Darwin (No
 DST)
 case 73: return "VUT-11"; // Vanuatu / New Caledonia (UTC+11)
 case 74: return "NFT-11"; // Norfolk Island, Australia (UTC+11 Fixed)
 case 75: return "CHUT-10"; // Chuuk / Micronesia (UTC+10)

For indexed timezones, the command has the following format:

```
[\\n] [11] [index] [\\r]
```

Example setting UK time:

```
Serial > \\n1123\\r
```

```
Serial < \\nOK\\r
```

For custom timezones, the command has the following format:

```
[\\n] [11] [-1] [posix format timezone string] [\\r]
```

Example setting UK time:

```
Serial > \\n11-1GMT0BST,M3.5.0/1,M10.5.0\\r
```

```
Serial < \\nOK\\r
```

Notes

The Software information of this datasheet is only valid for the Default code that is shipped in the device. The Default original code will always be available for reflash by the user.

Disclaimer:

No guarantees are given for fully compatibility with other codes.

The device hardware was designed to be compatible with the WLED software (and others) although this is only true at the moment of the device first version being released.

To program the device with WLED, all that is needed is to go to WLED page project and use the programming webpage. The whole process should be fully automatic.

The leds are ws2812.

The 2D matrix organisation configuration for WLED should be:

8 Tiles total, 1 vertical x 8 horizontal, organized from left to right.

Each tile starts from top left to bottom right in zigzag.



Others:

